

Pathmika Weeraratna

Associate Tech Lead | Full-Stack Engineer (Mobile & Web), AI/ML Exposure

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LinkedIn | GitHub

Summary

Full-stack engineer and Associate Tech Lead with nearly five years of experience delivering production mobile and web applications for telecom operators, having grown through every engineering level at Omobio. Contributed to 25+ client and product engagements – delivery, architecture, and pre-sales solutioning – across 9+ countries in Asia, Africa, and Latin America. Core strengths in React Native mobile development (native modules, app security, monetization) and Next.js/React web platforms, backed by solid REST API and backend engineering. Alongside this, gained hands-on AI/ML exposure self-initiating and leading a telecom fraud-detection R&D initiative spanning NLP embeddings, GenAI, and ensemble ML models. Deepening technical foundations through an MSc in Artificial Intelligence at the University of Moratuwa. Recognized as Emerging Employee of the Year (2023).

Experience

Associate Tech Lead

Jan 2026 – Present

Omobio (Pvt) Ltd

- Lead, mentor, and scale a cross-functional engineering team of 10+ developers, driving solution design and delivery across mobile, web, and data-platform initiatives.
- Serve as technical subject-matter expert for pre-sales engagements, producing architecture proposals and technical validation.
- Lead two flagship platforms: the AI/ML R&D team enhancing the SMS Firewall product with ML-driven detection, and the SOR (Steering of Roaming) centralized reporting platform.
- Leverage AI tools throughout the design and development workflow to boost personal productivity, enabling faster turnaround and higher-quality output.
- Manage client engagements end-to-end.

Senior Software Engineer

Jan 2024 – Jan 2026

Omobio (Pvt) Ltd

- Led Agile delivery of end-to-end mobile and web applications, owning architecture design and effort estimation.
- Delivered SQL/CDR reporting engineering for Banglalink's SMSC billing platform (Bangladesh), recovering ~85,000 dropped records and fixing a 240x query-performance regression.
- Architected cross-platform mobile solutions in React Native with deep native integration (Java, Kotlin, Swift), including a ~2,700-line custom native alarm engine built from the ground up across 26 files, replacing third-party alarm libraries on Android and iOS.
- Integrated RevenueCat for end-to-end in-app subscription purchasing.
- Self-initiated the AI/ML fraud-detection R&D work now powering the SMS Firewall AI/ML Suite (see Key Projects).
- Built HOPP, Hutch's Next.js 14 web self-care portal: authoring the core auth/locale middleware, a custom Axios API layer, and end-to-end integration across six payment methods.
- Continued ownership of Hutch Selfcare (led for 2+ years across this role and the one prior): led its React Native version upgrade from 0.66.5 to 0.76.5 and delivered 3+ change requests for new feature additions.
- Mentored a team of 10 junior developers, conducted code reviews, and established coding standards and GitHub Actions CI/CD pipelines to improve software reliability.

Software Engineer

Apr 2022 – Jan 2024

Omobio (Pvt) Ltd

- Led design and development of cross-platform mobile applications (Hutch Selfcare, Vodafone Selfcare, MyDialog, RetailHub, Circadian, SFA) in React Native, managing releases across Play Store, App Store, and Huawei AppGallery.
- Contributed to Dialog's enterprise web platforms (MDA, CDR-disclosure portal) using PHP Yii2 and React.
- Hardened Hutch Selfcare against fraud with certificate/public-key pinning and a device-integrity gate, and led the React Native version upgrade of Vodafone Fiji's production self-care app from RN 0.59.3 to 0.70.5.
- Participated in pre-sales engagement and contributed to ISO 27001 readiness documentation and a company-wide Git branching standard.
- Received the Emerging Employee of the Year Award (2023) for outstanding technical contribution.

- Supported frontend, backend, and mobile feature development for enterprise-grade projects using React.js, React Native, Python, and RESTful APIs; transitioned to a full-time Software Engineer role based on strong performance.

Key Projects

- **Hutch Selfcare (Telecom Self-Care App), 2+ years:** Owned and evolved Hutch's flagship React Native app: security hardening, dual FCM/HMS push architecture, a RN version upgrade (0.66.5 → 0.76.5), and 3+ feature change requests.
- **HOPP Web Self-Care Portal, 2024:** Next.js/TypeScript portal with custom Axios API layer and payment-gateway integration, delivered under a Hybrid Git Flow model.
- **SMS Firewall AI/ML Suite:** Lead a modular AI/ML detection suite (FastAPI, containerized modules, single admin CMS) covering OTP-bypass, phishing-URL, encoded-Unicode, grey-route, and campaign detection; an XGBoost/CatBoost ensemble powers phishing-URL detection, the OTP-bypass/grey-route and encoded-Unicode modules layer in semantic embeddings, and a local-LLM GenAI layer supports content review and translation across the suite.
- **SOR (Centralized Reporting Platform):** Lead a centralized telecom data-reporting platform spanning backend pipelines and web dashboards, with asynchronous task processing and containerized deployment.
- **Circadian (Native Alarm Engine & Monetization), 2024–2025:** Fully custom cross-platform alarm engine (reboot-persistence, background/killed-app handling, astronomy-driven scheduling) with RevenueCat-based subscription and entitlement management.
- **Robi Bangla SMS Opt-In/Opt-Out Platform, 2023:** Reviewed the SRS and contributed solution architecture for a BTRC-mandated language-preference platform (Bangladesh): a React/PHP Yii2/Erlang stack serving 5 user roles with sub-100ms REST APIs.

Additional Experience & Engagements

Separately architected FirewallBI, the large-scale telecom BI platform underpinning the AI/ML suite above (2023–2025; 8 on-prem sites, 300+ Logstash pipelines, mutual-TLS Elasticsearch cluster). Also led a major React Native version upgrade for Vodafone Fiji's production self-care app (2022, New Architecture scaffolding), and contributed to 15+ further engagements, including Dialog's CMDP, RetailHub, WinBack, and DF platforms; the RoamerSteering ELK analytics platform (2023); and solution proposals for Ncell (Nepal), Grameenphone (Bangladesh), and Ethio Telecom (Ethiopia).

Areas of Expertise

- **Mobile:** React Native (incl. New Architecture), Android (Java/Kotlin), iOS (Swift/Objective-C), Huawei HMS, native module bridging
- **Web:** React.js, Next.js, TypeScript/JavaScript, Redux/Redux-Saga
- **Backend & APIs:** Python, FastAPI, REST API design, Microservices, Java (Spring Boot), .NET (ASP.NET Core)
- **AI/ML (exposure):** PyTorch, Hugging Face Transformers, LangChain, XGBoost, CatBoost, SHAP, NLP, Semantic Search
- **Data Engineering:** PostgreSQL, ClickHouse, MariaDB/MySQL, Elasticsearch/ELK, Apache Airflow, Celery
- **DevOps:** Docker, Git, CI/CD (GitHub Actions, GitLab CI, Jenkins), Linux, Prometheus, Grafana
- **Leadership:** Agile delivery, technical mentoring, pre-sales & client engagement, architecture ownership

Education

Master of Science in Artificial Intelligence (Reading) – University of Moratuwa July 2025 – Present (Expected 2027)

B.Sc. in Information Systems, Second Class Upper Division, GPA 3.44 – University of Colombo School of Computing 2019 – 2022

Diploma in English (Certificate Level), Distinction Pass – Seagis Campus, Nugegoda

2018

Achievements

- Emerging Employee of the Year – Omobio (2023): recognized for exceptional job performance and contribution to key software engineering projects.
- Director's List Awardee – University of Colombo (2021): achieved GPA above 3.7 in first academic year.

References available upon request.